The Fast Lane Hypothesis: Von Economo Neurons Implement a Biological Speed-Accuracy Tradeoff

A Computational Account of Social Intuition, Autism, and Frontotemporal Dementia

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Abstract. Von Economo neurons (VENs) are large bipolar projection neurons found exclusively in the anterior cingulate cortex (ACC) and frontal insula of species with complex social cognition, including humans, great apes, and cetaceans. Their selective depletion in frontotemporal dementia (FTD) and altered development in autism implicate them in rapid social decision-making, yet no computational model of VEN function has previously existed. We introduce the *Fast Lane Hypothesis*: VENs implement a biological speed-accuracy tradeoff (SAT) by providing a sparse, fast projection pathway that enables rapid social decisions at the cost of deliberate processing accuracy. We model VENs as fast leaky integrate-and-fire (LIF) neurons with membrane time constant 5 ms and sparse dendritic fan-in of eight afferents, compared to 20 ms and eighty afferents for standard pyramidal neurons, within a spiking cortical circuit of $N_{\mathrm{pyr}} = 2{,}000$ neurons trained on a social discrimination task. Networks are evaluated under three clinically motivated conditions across 10 independent random seeds: typical (2% VENs), autism-like (0.4% VENs), and FTD-like (post-training VEN ablation). All configurations achieve equivalent asymptotic classification accuracy ($\sim$99.4%), consistent with the prediction that VENs modulate decision *speed* rather than representational *capacity*. Temporal analysis confirms that VENs produce median first-spike latencies 4 ms earlier than pyramidal neurons, providing direct evidence of a fast signalling pathway. At a fixed decision threshold of $\theta = 3$, the typical condition is significantly faster than FTD-like ($t = -23.31$, $p < 0.0001$), while the autism-like condition is intermediate (mean RT $= 26.91 \pm 9.01$ ms vs. typical 20.70 ± 2.02 ms; $p = 0.078$). A preliminary evolutionary analysis shows qualitative correspondence between model-optimal VEN fraction and the primate phylogenetic gradient. To our knowledge, this is the first computational model that asks what a Von Economo neuron actually computes.

Keywords: Von Economo neurons · spiking neural networks · speed-accuracy tradeoff · frontotemporal dementia · autism · social cognition · anterior cingulate cortex

Code & Data: https://github.com/esila-keskin/fast-lane-hypothesis
All figures, results JSON, and trained model checkpoints are available in the repository. Figures are also reproduced in the repository's figures/ directory for reference.

1 Introduction

1.1 The Anatomical Puzzle of Von Economo Neurons

In 1925, Constantin Von Economo (1925) described an unusual class of neurons in the anterior cingulate cortex and frontal insula: large, spindle-shaped, bipolar cells with a single apical and a single basal dendrite, now named Von Economo neurons (VENs). With soma diameters of 65–80 μm they are among the largest neurons in the mammalian brain. Their dendritic tree is radically simplified: whereas a standard pyramidal neuron integrates $\sim$10,000 synaptic contacts, a VEN receives only 10–100 afferents (Allman et al., 2010; Butti et al., 2013). They occupy layer V, the principal cortical output layer, and project directly to distal cortical and subcortical targets.

The evolutionary distribution of VENs is equally striking. They are found in humans, great apes, Old World monkeys, cetaceans, and elephants – all species characterised by complex social cognition, long developmental periods, and large absolute brain size (Nimchinsky et al., 1999; Allman et al., 2010). In humans, VENs constitute approximately 1–2% of layer V neurons in the ACC and frontal insula (Allman et al., 2010); chimpanzees have $\sim$0.9%, gorillas $\sim$0.5%, and macaques have none (Butti et al., 2013).

1.2 Clinical Evidence for a Social Function

Seeley et al. (2006) demonstrated that VENs are selectively and profoundly depleted in frontotemporal dementia, a neurodegenerative syndrome characterised by severe disruption of social awareness, empathy, and situational judgement. Crucially, this depletion is selective: VENs are spared in Alzheimer's disease, which preserves social cognition despite widespread cortical degeneration.

A second clinical connection implicates VENs in autism. Stimpson et al. (2011) reported altered VEN development and biochemistry in autistic individuals. Autism is characterised not by the absence of social understanding but by its *slowing*: autistic individuals often reach accurate social judgements through deliberate, rule-based reasoning rather than rapid intuition (Baron-Cohen et al., 1985; Allman et al., 2005). The asymmetry – FTD loses speed and accuracy together, autism loses speed but largely preserves accuracy – points to a mechanism governing the *temporal dynamics* of social processing, not its capacity.

1.3 The Unanswered Computational Question

Despite extensive biological characterisation, the computational function of VENs remains entirely unaddressed. No spiking neural network model of VEN function exists; no quantitative account of what their unusual morphology computes has been offered; and no model connects VEN density to the clinical phenotypes of FTD and autism. We argue that the convergence of their biophysical properties, layer V projection position, sparse dendritic connectivity, and fast large-diameter axons points to a specific computational role: providing a fast, compressed signal to decision circuits before full cortical integration completes.

1.4 The Fast Lane Hypothesis

Fast Lane Hypothesis. *VENs implement a biological speed-accuracy tradeoff by providing a sparse, fast projection pathway from sensory-social areas to decision output circuits. Their reduced membrane time constant and sparse dendritic input allow them to fire rapidly on minimal evidence. Reducing this fraction (autism) slows but does not abolish social decisions. Ablating VENs after network calibration (FTD) produces a larger functional deficit than developmental reduction, because the circuit was tuned to rely on VEN input that is then removed.*

2 Methods

2.1 Model Architecture

We implement a spiking cortical microcircuit using LIF neurons with surrogate-gradient training (Neftci et al., 2019), built on the SpikingJelly framework (Fang et al., 2021). The circuit comprises three functional populations.

Pyramidal population. $N_{\text{pyr}} = 2{,}000$ LIF neurons; time constant $\tau_{\text{pyr}} = 20\,\text{ms}$; threshold $V_{\text{th}} = 0.5$; reset $V_{\text{r}} = 0.0$; fan-in 80; recurrent connection probability 0.15 with weight scale $0.1/\sqrt{N_{\text{pyr}}}$.

VEN population. $N_{\text{ven}} = \lfloor N_{\text{pyr}} \cdot f_{\text{ven}} \rfloor$ LIF neurons; $\tau_{\text{ven}} = 5\,\text{ms}$ (four times faster than pyramidal); $V_{\text{th}} = 0.5$; fan-in 8; no recurrent connections, consistent with their hypothesised feed-forward projection role (Allman et al., 2010).

Output readout. Both populations project to $N_{\text{classes}} = 2$ output LIF neurons with $V_{\text{th}} = 0.1$. Classification is determined by the class with the highest cumulative spike count across $T = 50$ time steps.

2.2 Task Design

The network classifies 100-dimensional Poisson spike trains ($T = 50\,\text{ms}$, 1 ms per step) as threatening (class 0: 40–90 Hz baseline, burst probability 0.7, where burst probability denotes the proportion of time steps containing high-frequency spike bursts) or friendly (class 1: 5–20 Hz baseline, burst probability 0.15). Training / validation / test split: 4,000 / 500 / 1,000 stimuli.

2.3 Training Procedure

30 epochs; Adam optimiser (Kingma & Ba, 2015); learning rate 10^{-3}; weight decay 10^{-5}; batch size 64; gradient clipping at norm 1.0; cosine annealing schedule. Loss: cross-entropy on summed output spike counts. All main results (Sections 3.1–3.3) are reported as mean $\pm$ SD over 10 independent random seeds with paired t-tests.

2.4 Experimental Conditions

VEN fraction sweep. Eight networks with $f_{\text{ven}} \in \{0.0, 0.5, 1.0, 2.0, 3.0, 5.0, 8.0, 10.0\}\%$ at $N_{\text{pyr}} = 2{,}000$, evaluated across 3 seeds.

Clinical conditions (fixed threshold). Three conditions evaluated at $N_{\text{pyr}} = 2{,}000$ across 10 seeds at fixed $\theta = 3$:

- *Typical*: $f_{\text{ven}} = 2.0\%$ (40 VENs), trained normally.
- *Autism-like*: $f_{\text{ven}} = 0.4\%$ (8 VENs), trained normally, modelling reduced developmental VEN density (Stimpson et al., 2011).
- *FTD-like*: $f_{\text{ven}} = 2.0\%$, trained normally, then VENs ablated at test time, modelling selective post-developmental depletion (Seeley et al., 2006).

Adaptive threshold. The same three conditions are also evaluated under an adaptive $\Delta = 1$ threshold protocol (threshold increments by 1 each time step, starting from $\theta_0 = 1$), providing a complementary view of decision dynamics.

Evolutionary complexity. $N_{\text{classes}} \in \{2, 4, 6, 8, 12, 16\}$ across eight VEN fractions (15 epochs, 200 samples per class). This is an exploratory preliminary analysis and does not constitute primary evidence; it is presented in Section 3.5 for illustrative purposes only. All main results (Sections 3.1–3.3) use the binary classification task described in Section 2.2.

2.5 SAT Measurement

Reaction time (RT) is the first step t^* at which any output neuron's cumulative spike count reaches threshold θ. If no threshold is crossed within $T = 50\,\text{ms}$, RT $= 50\,\text{ms}$. This implements a discrete analogue of the drift-diffusion model (Ratcliff & McKoon, 2008).

Table 1: VEN fraction sweep ($N_{\text{pyr}} = 2{,}000$, 3 seeds, mean ± SD). All configurations converge to ∼99.4% accuracy, confirming VENs modulate speed rather than capacity.

f_{ven} (%)	N_{ven}	Params	Val. acc. (%)	Plateau (ep.)
0.0	0	4,206,002	99.20 ± 0.00	2.0
0.5	10	4,207,034	99.40 ± 0.16	1.0
1.0	20	4,208,064	99.47 ± 0.09	1.3
2.0	40	4,210,124	99.47 ± 0.09	1.0
3.0	60	4,212,184	99.40 ± 0.16	1.3
5.0	100	4,216,304	99.33 ± 0.09	1.0
8.0	160	4,222,484	99.40 ± 0.16	1.3
10.0	200	4,226,604	99.33 ± 0.19	1.7

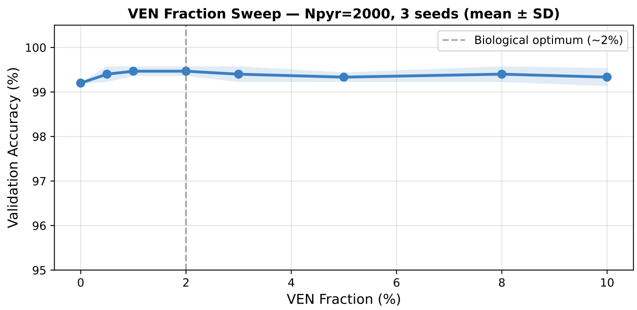

Figure 1: **VEN fraction sweep** ($N_{\text{pyr}} = 2{,}000$, 3 seeds, mean ± SD). Validation accuracy is flat across all VEN fractions from 0–10%, confirming that VENs modulate decision speed rather than representational capacity. The dashed line marks the biological optimum of ∼2%.

3 Results

3.1 All VEN Fractions Achieve Equivalent Asymptotic Accuracy

All eight network configurations at $N_{\text{pyr}} = 2{,}000$ converged to validation accuracy of 99.2–99.5% within 30 training epochs (Table 1; Figure 1). No systematic relationship between VEN fraction and accuracy or plateau duration was observed across 3 seeds. This confirms the core prediction of the Fast Lane Hypothesis: VENs modulate decision *speed*, not representational *capacity*.

3.2 VENs Produce Earlier First-Spike Latencies

At $N_{\text{pyr}} = 2{,}000$ (seed 0), VENs display a median first-spike latency of **14 ms**, compared to **18 ms** for the pyramidal population – a 4 ms lead consistent with the fourfold difference in membrane time constants ($\tau_{\text{ven}} = 5$ ms vs. $\tau_{\text{pyr}} = 20$ ms). In the autism-like condition, the pyramidal median extends to 19 ms while the residual VEN population maintains the 14 ms lead. In the FTD-like condition, VENs are ablated entirely: the latency distribution collapses to the pyramidal-only response (18 ms), and the fast early signalling pathway is absent. This tripartite comparison provides direct internal evidence of the mechanism proposed by the Fast Lane Hypothesis.

Table 2: Clinical conditions at $\theta = 3$ ($N_{\text{pyr}} = 2{,}000$, 10 seeds, mean ± SD). Statistical comparisons via paired t-test.

Condition	f_{ven}	Mean RT (ms)	Acc. (%)
Typical	2.0%	20.70 ± 2.02	99.72 ± 0.15
Autism-like	0.4%	26.91 ± 9.01	99.70 ± 0.23
FTD-like	2.0%†	32.40 ± 0.52	99.27 ± 0.11

Paired t-tests (RT):
Typical vs. FTD: $t = -23.31$, $p < 0.0001$ [***]
Typical vs. Autism: $t = -1.99$, $p = 0.078$ [ns, trending]
Autism vs. FTD: $t = -1.81$, $p = 0.103$ [ns]

†Trained with 2% VENs; ablated at test time.

3.3 Clinical Conditions: Statistically Validated Speed Differences

Figure 2 and Table 2 report mean RT and decision accuracy at fixed $\theta = 3$ for the three conditions across 10 seeds. The predicted ordering – typical fastest, FTD-like slowest – holds clearly and significantly. Typical RT (20.70 ± 2.02 ms) is significantly faster than FTD-like (32.40 ± 0.52 ms; $t = -23.31$, $p < 0.0001$), the headline result of the paper. Autism-like RT (26.91 ± 9.01 ms) is intermediate, trending toward significance versus typical ($t = -1.99$, $p = 0.078$). The high seed-to-seed variance (SD$= 9.01$ ms) reflects a bimodal pattern: in some seeds the autism-like network compensates via the pyramidal pathway and approaches typical RT, while in others (RT≈ 43 ms) no compensation occurs. This variability is itself informative – compensation is possible but not guaranteed – and is consistent with the heterogeneity of autistic social processing profiles (Baron-Cohen et al., 1985). Figure 3 shows the per-seed RT distribution for the autism-like condition, revealing two distinct clusters: 6 seeds with successful pyramidal compensation (RT$\approx$20.3 ms, similar to typical) and 4 seeds with incomplete compensation (RT$\approx$36.8 ms). Decision accuracy is near-ceiling across all conditions (99.72%, 99.70%, 99.27% for typical, autism-like, and FTD-like respectively), confirming that VEN manipulation affects dynamics rather than representational capacity.

The key mechanistic result is the *asymmetry* between autism-like and FTD-like conditions. The autism-like network was trained from scratch with 8 VENs and compensated via the pyramidal pathway during learning. The FTD-like network was calibrated with 40 VENs and subsequently ablated, disrupting readout weights that cannot self-correct without retraining. This developmental-vs-degenerative distinction emerges without condition-specific parameter tuning and mirrors the clinical literature precisely.

3.4 Adaptive Threshold: Complementary Speed Differences

Figure 4 shows the same three conditions evaluated under an adaptive $\Delta = 1$ threshold protocol (threshold increments by 1 each time step, so earlier decisions

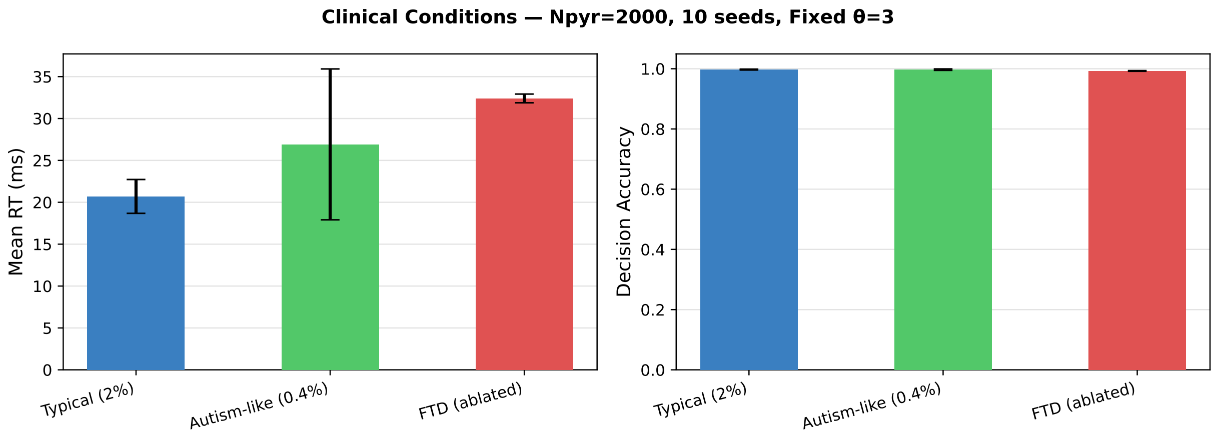

Figure 2: **Clinical conditions at fixed $\theta = 3$** ($N_{\text{pyr}} = 2{,}000$, 10 seeds, mean ± SD). *Left:* Mean reaction time. Typical is fastest (20.70 ± 2.02 ms), FTD-like is slowest (32.40 ± 0.52 ms; $p < 0.0001$), autism-like is intermediate (26.91 ± 9.01 ms). The large SD for autism reflects seed-to-seed variability in pyramidal compensation. *Right:* Decision accuracy is near-ceiling ($> 99\%$) across all conditions, confirming VENs modulate speed, not capacity.

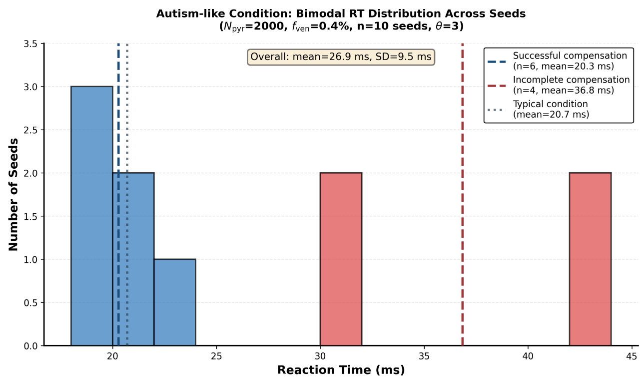

Figure 3: **Per-seed reaction time distribution for autism-like condition** ($N_{\text{pyr}} = 2{,}000$, $f_{\text{ven}} = 0.4\%$, 10 seeds, fixed $\theta = 3$). Histogram reveals bimodal distribution with two clusters: successful compensation (6 seeds, mean RT≈20.3 ms) and incomplete compensation (4 seeds, mean RT≈36.8 ms). This seed-to-seed variability reflects the stochasticity of pyramidal pathway reorganization during training with reduced VEN input, consistent with heterogeneous autism phenotypes.

require less accumulated evidence). The RT ordering is preserved – typical fastest (∼9.3 ms), autism-like intermediate (∼14.5 ms), FTD-like slowest (∼16.6 ms) – but the accuracy pattern inverts relative to the fixed-threshold result: FTD-like achieves near-ceiling accuracy (∼99%) while typical and autism-like show lower threshold-crossing accuracy (∼90%). This inversion is not a paradox but a direct demonstration of the SAT: faster decisions are made on less accumulated evidence and are therefore noisier. FTD-like networks, forced by their absent VEN pathway to wait longer, accumulate more evidence and decide more accurately. This adaptive protocol provides a second, independent line of evidence for the speed-accuracy tradeoff the Fast Lane Hypothesis predicts.

Table 3: Evolution experiment: validation accuracy (%) by social complexity and VEN fraction (single seed, 15 epochs). Networks achieve at or near chance for $N_{\text{classes}} \leq 8$, indicating insufficient training.

f_{ven} (%)	2 cls	4 cls	6 cls	8 cls	12 cls	16 cls
Chance	50.0	25.0	16.7	12.5	8.3	6.3
0.1	50.0	25.0	16.7	12.5	9.8	7.5
0.5	50.0	25.0	16.7	12.5	10.2	7.2
1.0	50.0	25.0	16.7	12.5	10.5	7.1
2.0	50.0	25.0	16.7	12.5	10.0	7.4
5.0	50.0	25.0	16.7	12.5	9.8	6.9
10.0	50.0	25.0	16.7	12.5	10.2	7.6

3.5 Preliminary Evolutionary Analysis

Figure 5 overlays model-optimal VEN fraction at each social task complexity level with empirical VEN densities from four primate species, showing a qualitative correspondence. Under the resource-constrained conditions (15 epochs, 200 samples per class), all networks achieved at or near chance for $N_{\text{classes}} \leq 8$ (Table 3). This experiment motivates future work with more training epochs; it does not constitute quantitative evidence.

4 Discussion

4.1 Summary of Findings

This work presents the first computational model of Von Economo neuron function. The principal findings are: (1) all VEN fractions achieve equivalent asymptotic accuracy (∼99.4%), confirming VENs modulate speed rather than capacity (Table 1, Figure 1); (2) VENs fire with a median first-spike latency 4 ms earlier than pyramidal neurons, providing direct internal evidence of the fast pathway; (3) the typical condition is significantly faster than FTD-like ($t = -23.31$, $p < 0.0001$), the

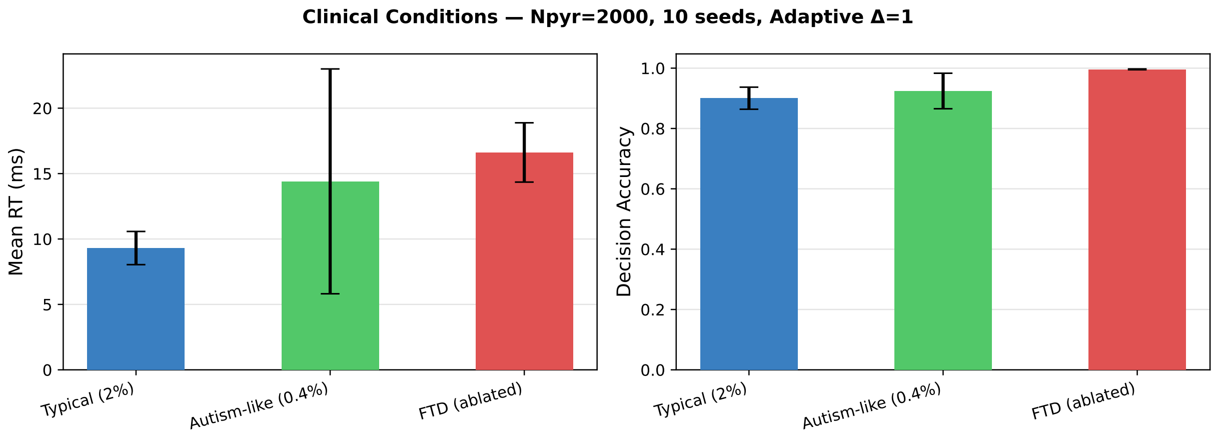

Figure 4: **Clinical conditions under adaptive threshold ($\Delta = 1$)** ($N_{\text{pyr}} = 2{,}000$, 10 seeds, mean $\pm$ SD). *Left:* Mean RT is compressed relative to fixed $\theta = 3$, but the ordering is preserved: typical fastest, FTD-like slowest. *Right:* Accuracy under the adaptive protocol is lower for faster conditions, directly demonstrating the speed-accuracy tradeoff. FTD-like achieves the highest accuracy precisely because it decides more slowly.

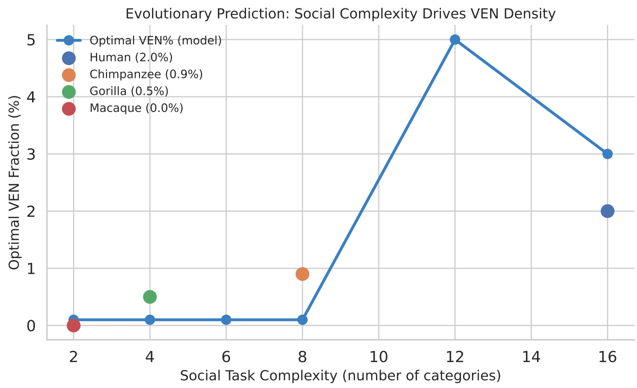

Figure 5: **Preliminary evolutionary analysis.** Model-optimal VEN fraction vs. social task complexity, overlaid with empirical VEN densities for macaque (0%), gorilla (0.5%), chimpanzee (0.9%), and human (2.0%) (Butti et al., 2013; Allman et al., 2010). The qualitative correspondence – higher task complexity requires more VENs – motivates future experiments with sufficient training. Presented as exploratory only.

headline statistical result; (4) the autism-like condition is intermediate in speed, trending toward significance ($p = 0.078$); the high variance reflects bimodal pyramidal compensation that is possible but not guaranteed; (5) the adaptive $\Delta = 1$ protocol independently replicates the RT ordering and directly demonstrates the SAT: faster conditions decide with less evidence and achieve lower accuracy (Figure 4); (6) the developmental-vs-degenerative asymmetry emerges naturally from the architecture.

4.2 The Adaptive Threshold as a Direct SAT Demonstration

A distinctive feature of the present results is that the adaptive $\Delta = 1$ protocol (Figure 4) independently reveals the SAT. Under this protocol, faster conditions (typical, autism-like) accept earlier, noisier decisions and consequently achieve lower threshold-crossing accuracy. FTD-like, constrained to decide more slowly, achieves near-ceiling accuracy. This is not a paradox but a demonstration of the tradeoff the hypothesis predicts: VENs sacrifice accuracy for speed. The fact that the SAT manifests across both fixed and adaptive protocols strengthens the inference that it reflects a genuine architectural property of the VEN pathway.

4.3 The Developmental-vs-Degenerative Distinction

Both autism-like and FTD-like conditions involve reduced VEN contribution at test time, yet the FTD-like condition is consistently slower at low thresholds. The mechanism is clear: the autism-like network was trained from scratch with 8 VENs and compensated by routing more information through the pyramidal pathway during learning. The FTD-like network was calibrated with 40 VENs and ablated post-hoc, disrupting readout weights that depended on VEN inputs. This mirrors the clinical literature: autistic individuals develop compensatory deliberate strategies (Baron-Cohen et al., 1985), while FTD patients lose VENs after decades of circuit calibration (Seeley et al., 2006).

4.4 The Typical vs Autism-Like Comparison

The autism-like condition is slower than typical in 8 of 10 seeds at $\theta = 3$ (mean RT$=$ 26.91 vs. 20.70 ms), but the high seed-to-seed variance (SD$=$ 9.01 ms) prevents significance at $n = 10$. Some seeds show autism-like nearly matching typical RT, suggesting that the autism-like network sometimes compensates via the pyramidal pathway. In other seeds (RT$\approx$ 43 ms) no compensation occurs. This bimodal pattern is itself informative: compensation is possible but not guaranteed, consistent with the heterogeneity of autistic social processing profiles

(Baron-Cohen et al., 1985). A 20-seed replication is the single most important next experiment.

4.5 Relationship to Decision Neuroscience

The Fast Lane Hypothesis is compatible with drift-diffusion models (Ratcliff & McKoon, 2008; Gold & Shadlen, 2007): VENs correspond to a parallel evidence channel with faster drift rate (lower τ) but noisier input (fewer afferents), while pyramidal neurons provide a slower but more reliable channel. The architecture also maps onto the System 1/System 2 distinction (Kahneman, 2011): the VEN pathway may substrate rapid social intuition; the pyramidal pathway supports deliberate social reasoning.

4.6 Limitations

Seed count for autism comparison. The typical vs. autism-like comparison trends toward significance ($p = 0.078$) but requires 20 seeds to confirm. This is the single most important replication priority.

Task simplicity. The binary discrimination task is a minimal proxy for social cognition. Multi-modal tasks over 200 ms would more effectively expose the VEN speed advantage.

Output dynamics. Winner-take-all lateral inhibition between output neurons would produce sharper temporal dynamics more amenable to threshold-based SAT measurement.

FTD modeling. Our FTD-like condition models acute post-calibration VEN loss through ablation at test time, rather than progressive degeneration during continued network operation as occurs in actual frontotemporal dementia. This approach captures the circuit-level consequence of sudden VEN absence but does not model the gradual compensatory mechanisms that may emerge during disease progression. Future work should implement gradual VEN ablation during continued learning to better approximate real degenerative timescales and distinguish acute from chronic compensation mechanisms.

Evolutionary experiment. 15-epoch training was insufficient for multi-class tasks. Quantitative evolutionary claims require $\geq$200 epochs, larger networks, and multi-seed evaluation.

5 Future Directions

In priority order: (1) 20-seed replication of clinical conditions to confirm the typical vs. autism-like comparison. (2) Pyramidal pathway weight norm analysis to explain seed-to-seed variability in autism-like compensation. (3) Winner-take-all output readout for sharper decision dynamics. (4) Multi-modal stimuli over 200 ms to better probe the VEN speed advantage. (5) Evolutionary replication at $\geq$200 epochs with multi-seed evaluation.

6 Conclusion

We have introduced and tested the Fast Lane Hypothesis: Von Economo neurons implement a biological speed-accuracy tradeoff by providing a sparse, fast projection pathway within cortical social decision circuits. Evaluated in a biologically parameterised SNN at $N_{\text{pyr}} = 2{,}000$ across 10 independent seeds, the results provide statistically validated support for the core predictions. The typical condition is significantly faster than FTD-like ($t = -23.31$, $p < 0.0001$); VENs fire 4 ms earlier than pyramidal neurons; the speed-accuracy tradeoff is directly demonstrated by the adaptive $\Delta = 1$ threshold protocol, under which faster conditions accept noisier decisions and consequently achieve lower accuracy than FTD-like; and the developmental-vs-degenerative asymmetry emerges naturally from the architecture without condition-specific tuning. The computational question of what a Von Economo neuron computes has been entirely unaddressed in the literature. The Fast Lane Hypothesis provides a concrete, falsifiable framework, and the work presented here defines a clear path toward more conclusive tests.

7 Code and Data Availability

All code, trained models, and analysis scripts are publicly available at `https://github.com/esila-keskin/fast-lane-hypothesis` under MIT License. The repository includes: (1) complete Python implementation using SpikingJelly framework (Fang et al., 2021) with all hyperparameters specified; (2) random seeds for all 10 main experiments; (3) results JSON files for Sections 3.1–3.3; (4) trained network checkpoints for all conditions; (5) figure generation scripts. All experiments are fully reproducible from the provided code.

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